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# Appendix

## A Datasets and Implementations

### A.1 Selected Relations and Prompt Templates

We employed a set of criteria to select relations from Wikidata in order to construct our dataset. Our criteria largely follow the framework proposed by Gekhman et al. (2025). Specifically, we require that each factual query in the dataset be unambiguous: given a subject–relation pair, the object should be unique and easy verifiable. The criteria are as follows:

- **Avoid granularity ambiguity**. We exclude relations whose answers can vary in their level of detail. For example, in *location* queries, the response could be expressed as a city, state, or country, making it ill-defined (Yona et al., 2024).

| Relation | Prompt Template |
|---|---|
| father | The name of the father of `[subject]` is |
| mother | The name of the mother of `[subject]` is |
| spouse | The name of the spouse of `[subject]` is |
| date of birth | The birth date of `[subject]` is |

Table 3: Relations and prompt templates for querying factual knowledge of models. `[subject]` is a placeholder replaced with subject entities.

- **Avoid surface-level guessing.** We exclude relations whose correct answers can often be inferred from shallow patterns. For instance, *country of citizenship* can frequently be guessed from shallow lexical patterns, rather then reflecting actual memorization (Mallen et al., 2023b).

Following these criteria, Gekhman et al. (2025) narrowed the 24 relations introduced by Sciavolino et al. (2021) down to four. However, we observe that their filtering primarily addresses ambiguity at the relation and object levels, but does not consider ambiguity at the subject level. In practice, some relations involve subjects that are inherently ambiguous. For example, the relation *record label* can be problematic because many songs share identical names, leading to unclear subject–object mappings.

To mitigate such cases, we apply an additional subject-level filtering step and restrict our dataset to relations where the subject is a person, thereby reducing ambiguity. In addition, we manually include certain relations to strengthen the dataset. Concretely, we use the following four relations: P22 (*father*), P25 (*mother*), P26 (*spouse*), and P569 (*date of birth*). We show the list of the templates used to create our dataset in Table 3.

### A.2 Implementation Details

**Checkpoints and GPU resources.** All the checkpoints used in our experiments are provided by the Hugging Face Transformers library (Wolf et al., 2019). Specifically, we use the checkpoint "meta-llama/Meta-Llama-3-8B"[1] and "mistralai/Mistral-7B-v0.3"[2] for the experiments of response generation (§3), hidden-state analysis (§4) and accessing the performance of hallucination detection methods (§5). For refusal tuning (§6), we use checkpoints

[1] https://huggingface.co/meta-llama/Meta-Llama-3-8B

[2] https://huggingface.co/mistralai/Mistral-7B-v0.3

**LLM Judge Prompt**

I will give you a factual query (e.g., "The name of the father of <subj>"), a gold answer to the factual query, and a proposed answer generated by an LLM. You need to compare the proposed answer to the gold answer and assign it one of the possible grades using the steps below.

**Possible grades are:**

A: CORRECT
B: INCORRECT
C: WRONG GOLD
D: ERROR

Spelling errors, synonyms, abbreviations, or hedging expressions (e.g., "it is possible that") should not alter the grade if the person referred to in the proposed answer matches the gold answer.

**Steps:**

Step 1: If the gold answer does not correspond to an answer for the question, output "C" and finish. Otherwise, proceed to Step 2.
Step 2: Extract all predicted entities from the proposed answer. Proceed to Step 3.
Step 3: If each predicted entity refers to the answer mentioned in the gold answer, output "A" and finish. Otherwise, proceed to Step 4.
Step 4: If the predicted entity does not refer to the gold answer, output "B" and finish. Otherwise, proceed to Step 5.
Step 5: Double-check whether the proposed answer refers to a different answer from the gold answer. If it does, output "B." Otherwise, output "D" and finish.

**Input format:**

Question: {question}
Gold answer: {gold_answer}
Proposed answer: {proposed_answer}

**Instruction:** Output your reasoning steps. After that, conclude your response with "Output:" followed by the letter (A, B, C, or D). Do not provide any further explanation.

Figure 11: LLM Judge prompt used for evaluation.

provided by the Unsloth framework (Daniel Han and team, 2023), namely "unsloth/llama-3-8b"[3] and "unsloth/mistral-7b-v0.3"[4], which enable more efficient fine-tuning. All experiments are conducted on 4 NVIDIA L40S GPUs.

**Decoding algorithm.** We employ greedy decoding (temperature $= 0$) for response generation, with models run in BF16 precision.

**PEFT settings for refusal tuning.** For refusal tuning, we fine-tune with both models using QLoRA (Dettmers et al., 2023), implemented with the Unsloth framework (Daniel Han and team, 2023), with rank $r = 8$, and $\alpha = 8$. QLoRA adapters are applied to all attention and MLP modules, and each model is fine-tuned for one epoch.

[3] https://huggingface.co/unsloth/llama-3-8b
[4] https://huggingface.co/unsloth/mistral-7b-v0.3

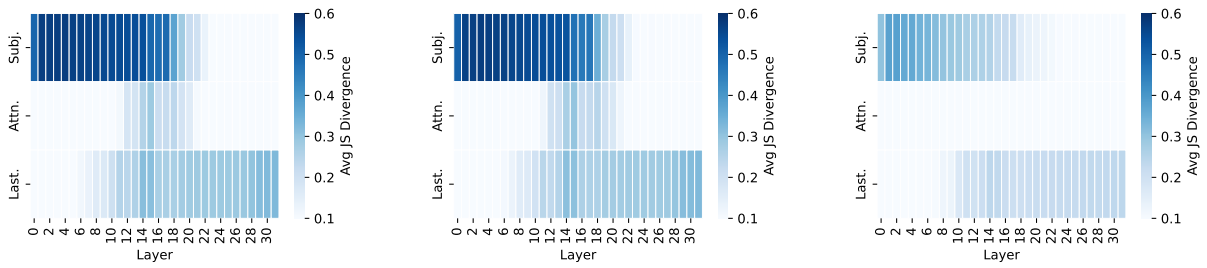
### A.3 Labeling Factual Correctness

Here, we outline the process of determine Factual Correctness of a model prediction. We construct correctness labels through a two-stage process. First, we use spaCy[5] Named Entity Recognizer to extract the target entity from the model's output. If it matches the ground truth, the answer is marked correct. Otherwise, or if extraction fails, we rely on Qwen2.5-14B-Instruct (Yang et al., 2024a) as a judge to compare the predicted answer with the ground truth. Following Gekhman et al. (2025), we design the evaluation prompt, which is shown in Figure 11.

## B Calculation of Subject Representation Norms

To investigate the geometric properties of subject representations, we compute the average $L_2$ norm

[5] https://spacy.io/




(a) Factual Associations (b) Associated Hallucinations (c) Unassociated Hallucinations

Figure 12: Effect of interventions across layers of Mistral-7B-v0.3. The heatmap shows JS divergence between the output distribution before and after intervention. Darker color indicates that the intervened hidden states are more causally influential on the model's predictions. Top row: patching representations of subject tokens. Middle row: blocking attention flow from subject to the last token. Bottom row: patching representations of the last token.

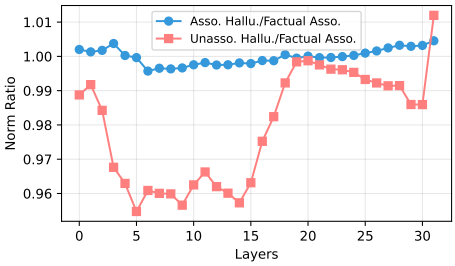



Figure 13: Norm ratio curves of subject representations in Mistral-7B-v0.3, comparing AHs and UHs against FAs as the baseline. At earlier layers, the norm of UH samples is significantly lower than that of AH samples.

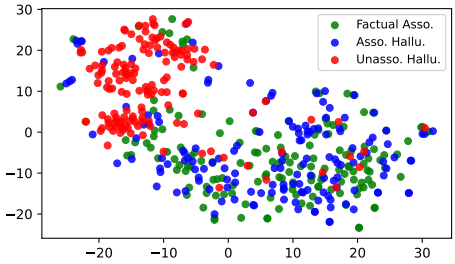



Figure 14: t-SNE visualization of last token's representations at layer 25 of Mistral-7B-v0.3.

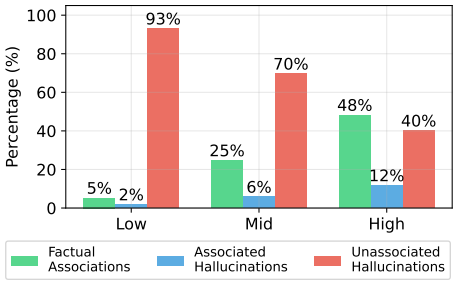



Figure 15: Sample distribution across different subject popularity (low, mid, high) in Mistral-7B-v0.3, measured by monthly Wikipedia page views.

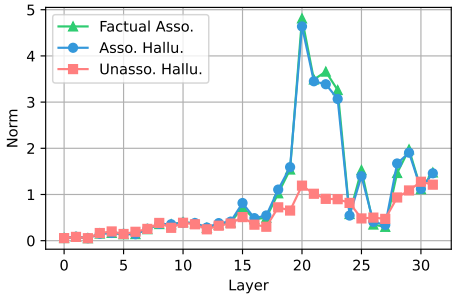



Figure 16: Subject-to-last attention contribution norms across layers in Mistral-7B-v0.3. Values show the norm of the attention contribution from subject tokens to the last token at each layer.

of the hidden states across different knowledge categories. For a single sample with subject tokens $s_1, \ldots, s_n$, we first compute the average norm across all subject tokens at layer $\ell$:

$$\bar{N}^{(\ell)}_{\text{sample}} = \frac{1}{n} \sum_{i=1}^{n} \left\| \mathbf{h}^{(\ell)}_{s_i} \right\|_2 \tag{2}$$

Then, for a given category (e.g., FAs, UHs, or AHs) containing $K$ samples, we aggregate these values to obtain the per-category average norm:

$$\bar{N}^{(\ell)}_{\text{category}} = \frac{1}{K} \sum_{j=1}^{K} \bar{N}^{(\ell)}_{\text{sample},j} \tag{3}$$

where $\bar{N}^{(\ell)}\text{sample}, j$ is the average norm for the $j$-th sample in the category. By repeating this process for each layer $\ell \in \{1, \ldots, L\}$, we obtain a layer-wise norm profile. Finally, to facilitate direct comparison, we report the norm ratio of hallucinations relative to factual associations:

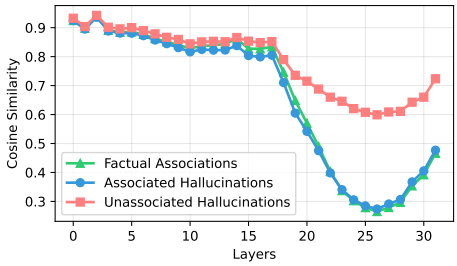



Figure 17: Cosine similarity of target-token hidden states across layers in Mistral-7B-v0.3. From mid-layers onward, FAs and AHs diverge sharply as subject information propagates, while UHs remain more clustered, confirming weaker subject-dependent updates.

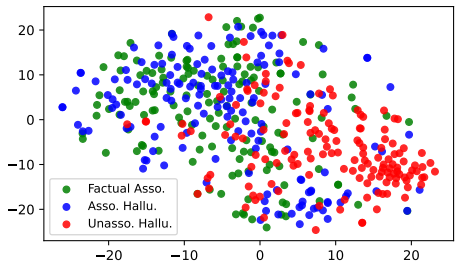



Figure 18: t-SNE visualization of subject tokens' representations at layer 11 of LLaMA-3-8B.

$$\text{Ratio}^{(\ell)}_{\text{AH/FA}} = \frac{\bar{N}^{(\ell)}_{\text{AH}}}{\bar{N}^{(\ell)}_{\text{FA}}} \tag{4}$$

$$\text{Ratio}^{(\ell)}_{\text{UH/FA}} = \frac{\bar{N}^{(\ell)}_{\text{UH}}}{\bar{N}^{(\ell)}_{\text{FA}}} \tag{5}$$

## C Parallel Experiments on Mistral

This section is for documenting parallel experiments conducted on the Mistral-7B-v0.3 model under the same settings as described in the main text (Figures 12–21).

The results from Mistral exhibit similar patterns to those observed in LLaMA, as described before. Specifically, we find consistent patterns in the model's internal computations, hidden-state behaviors, and the performance of hallucination detection and refusal tuning experiments.

## D Further Analysis on Internal States

### D.1 Analysis of Parametric Subspace Overlap for Subject Representation

We investigate why early layers encode subject representations differently across knowledge types by examining how inputs interact with the parametric

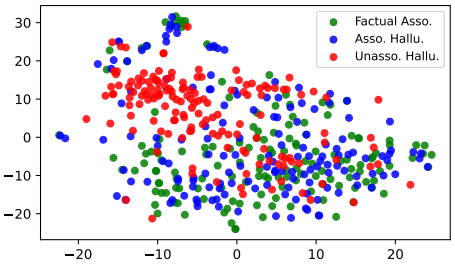



Figure 19: t-SNE visualization of subject tokens' representations at layer 11 of Mistral-7B-v0.3.

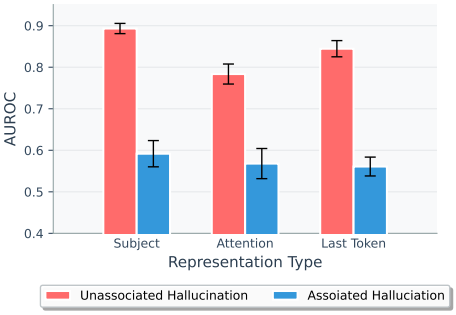



Figure 20: Hallucination detection performance on the **Full** setting (Mistral-7B-v0.3).

knowledge stored in MLP modules. Inspired by Kang et al. (2024), the output norm of an MLP layer depends on how well its input aligns with the subspace spanned by the weight matrix: *poorly aligned inputs yield smaller output norms*.

For each MLP layer $\ell$, we analyze the down-projection weight matrix $W^{\ell}_{\text{down}}$ and its input $x^{\ell}$. Given the input $x^{\ell}_s$ corresponding to the subject tokens, we compute its overlap ratio with the top singular subspace $V_{\text{top}}$ of $W^{\ell}_{\text{down}}$:

$$r(x^{\ell}_s) = \frac{\left\| x^{\ell\top}_s V_{\text{top}} V^{\top}_{\text{top}} \right\|^2}{\|x^{\ell}_s\|^2}. \tag{6}$$

A higher overlap ratio $r(x^{\ell}_s)$ indicates stronger alignment to the subspace spanned by $W^{\ell}_{\text{down}}$, leading to larger output norms.

To highlight relative deviations from the factual baseline (FA), we report the relative ratios between AH/FA and UH/FA. Focusing on the layer with the largest UH norm shift, Figure 22 shows that UHs have significantly lower $r(x^{\ell}_s)$ than AHs in both LLaMA and Mistral. This reveals that early-layer parametric weights are more aligned with FA and AH subject representations than with UH subjects, producing higher norms for the former ones. These results also suggest that the model has sufficiently learned representations for FA and AH subjects during pretraining but not for UH subjects.